



Self-Supervised Contrastive Learning for Automated Segmentation of Brain Tumor MRI Images in Schizophrenia

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Abstract

Schizophrenic patients' brain tumor magnetic resonance imaging (MRI) images are important references for doctors to diagnose and treat schizophrenia. However, automatic segmentation of these images is a professional and tedious task. Existing methods suffer from problems such as large model parameters, long computation time, and inadequate image processing. To achieve more accurate segmentation of brain tumors, we propose brain tumor MRI images for automatic segmentation using self-supervised contrastive learning in schizophrenia patients (BTCSSSP). First, a denoising algorithm based on progressive principal component analysis approximation and adaptive clustering is designed to process the noisy MRI images. Second, a brightness-aware image enhancement algorithm is developed to address the problems of non-uniformity, unclear boundaries, and poor spatial resolution of the MRI images. Finally, a cross-scale U-Net network with selective feature fusion attention module is designed based on self-supervised contrastive learning to achieve automatic segmentation of brain tumor MRI images. The results show that the BTCSSSP method yields higher Recall and Precision than existing methods. The maximum recall is 95%, and the image segmentation precision is 95%, thus indicating good practical applicability.

Keywords Brain tumor MRI images · Automatic segmentation · Schizophrenia patients · Self-supervised contrastive learning · Selective feature fusion attention module

1 Introduction

In recent years, significant progress has been made in the field of mental and neurological disorders. Depression, as a common mental health issue, has seen a surge in research activity. Applications of traditional and medical intelligence technologies, through systematic mapping and literature reviews, have analyzed 60 related articles, revealing associations between factors such as genetics and oxidative

stress with depression [1]. This underscores the importance of interdisciplinary approaches in treating depression and provides new perspectives for understanding the complexity of mental illnesses. At the same time, the prevalence of anxiety among university students has been steadily increasing, attracting widespread attention from both society and academia. A study utilizing an integrated framework of intelligent medical Internet-of-Things and Manosakti Yoga, and employing machine learning algorithms to analyze cardiac signal data, found that MSY effectively reduces anxiety and enhances cognitive function [2]. This discovery not only offers a new solution for alleviating anxiety among university students but also demonstrates the potential of intelligent technology in the field of mental health. Despite the severe threat that mental illnesses such as depression and anxiety pose to individual health, the diagnosis and treatment of brain tumors are equally critical [3]. Although the incidence of brain tumors is relatively low, their potential impact on health is significant. Particularly in patients with schizophrenia, the presence of brain tumors may exacerbate the condition, making early diagnosis and treatment crucial [4–6]. Magnetic resonance imaging (MRI) technology

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plays a vital role in detecting brain tumors and providing pathological information [7, 8]. However, MRI images often suffer from noise, inhomogeneity, unclear boundaries, and poor spatial resolution, making tumor segmentation a highly challenging task [9, 10]. To address the issues of excessive model parameters, long computation time, and inadequate image processing leading to suboptimal segmentation results in the aforementioned methods, we propose a self-supervised contrastive learning-based method for automatic segmentation of brain tumor MRI images in schizophrenia. The main contributions of this paper are as follows: (1) by denoising and enhancing MRI images of brain tumors, this paper significantly improves image quality, thereby facilitating the segmentation algorithm's ability to more effectively extract key information, further enhancing segmentation precision. (2) This paper incorporates a selective feature fusion attention module into the UNet architecture, enabling the network to identify more precise and discriminative feature representations, effectively preventing inaccuracies in segmentation due to insufficient image processing. (3) Through the introduction of skip connection strategies, this paper achieves direct communication between the encoder and decoder, promoting the effective integration of low-level and high-level features, effectively mitigating the challenges of excessive model parameters and prolonged computation time.

2 Related Works

Currently, research on MRI image segmentation has achieved significant progress. Scholars have applied techniques such as deep learning and machine learning [11], active contour models [12], region-based segmentation methods, and threshold segmentation methods [13] to achieve automatic segmentation of MRI images. Hao et al. [14] proposes an automatic segmentation method for the total target volume of nasopharyngeal carcinoma based on a spatial attention-based multiscale deep neural network ensemble. It utilizes a designed 2.5D convolutional neural networks (CNN) to handle the differences between two MRI imaging modalities and introduces a spatial attention module to focus on small targets, leveraging channel attention to further enhance segmentation performance. However, the 2.5D convolution operation, which processes different depth image slices simultaneously, may lead to unsatisfactory segmentation results if not properly handled. Li et al. [15] presented a U-Net-based method for cardiac MRI structure segmentation using multiscale features. It aggregates features at multiple levels and employs an attention mechanism to determine which features to focus on. The

decoder block also employs cascaded and parallel dilated convolutions, along with skip connections, to enhance the receptive field's capability. However, the multiscale features and attention mechanism increase the model's parameter count. Chatterjee et al. [16] proposed an automatic brain MRI segmentation method, utilizing the advantages of the classical random walker algorithm and graph cut optimization technique within a single framework. However, the random walker algorithm requires constructing a graph model and performing graph cuts, resulting in longer processing times. Veeramalla et al. [17] introduced an effective brain MRI segmentation algorithm based on a neuro-mechanical model. It combines statistical and partial differential equation-based methods to handle complex brain structures and different lesion types. By combining various segmentation methods, it can rapidly segment brain MRI data with the desired accuracy. However, the neuro-mechanical model and statistical methods may lead to increased model parameters and longer computation times when dealing with complex brain structures and different lesion types. Barzegar and Jamzad [18] proposed an automatic segmentation method for glioblastoma tumors in multimodal brain MRI using anatomical symmetric plane detection. It first detects symmetric planes and then defines a similarity measure for comparing the hemispheres using features extracted from both sides of the brain. The similarity measure is used to extract the tumor region, and post-processing operations are employed to improve accuracy. However, the detection of anatomical symmetric planes may require special handling and introduce additional computation time during the segmentation process.

3 Materials and Methods

This paper proposes a BTCSSSP method. The BTCSSSP method estimates the Gaussian noise level in the images using standard PCA and extracts image features using adaptive clustering and Multi-Scale Context Feature Blocks (MSCFB) technology. It trains the network model to learn feature representations through self-supervised contrastive learning, enhances feature attention with a selective feature fusion attention module, reduces model parameters and computational time through skip connection strategy, extracts feature with the encoder module, restores image resolution with the decoder module, and extracts multi-scale features using spatial pyramid pooling (SPP). The multi-scale output module obtains the final probability map, ultimately achieving the automatic segmentation task. The MRI image automatic segmentation framework based on self-supervised contrastive learning is illustrated in Fig. 1.

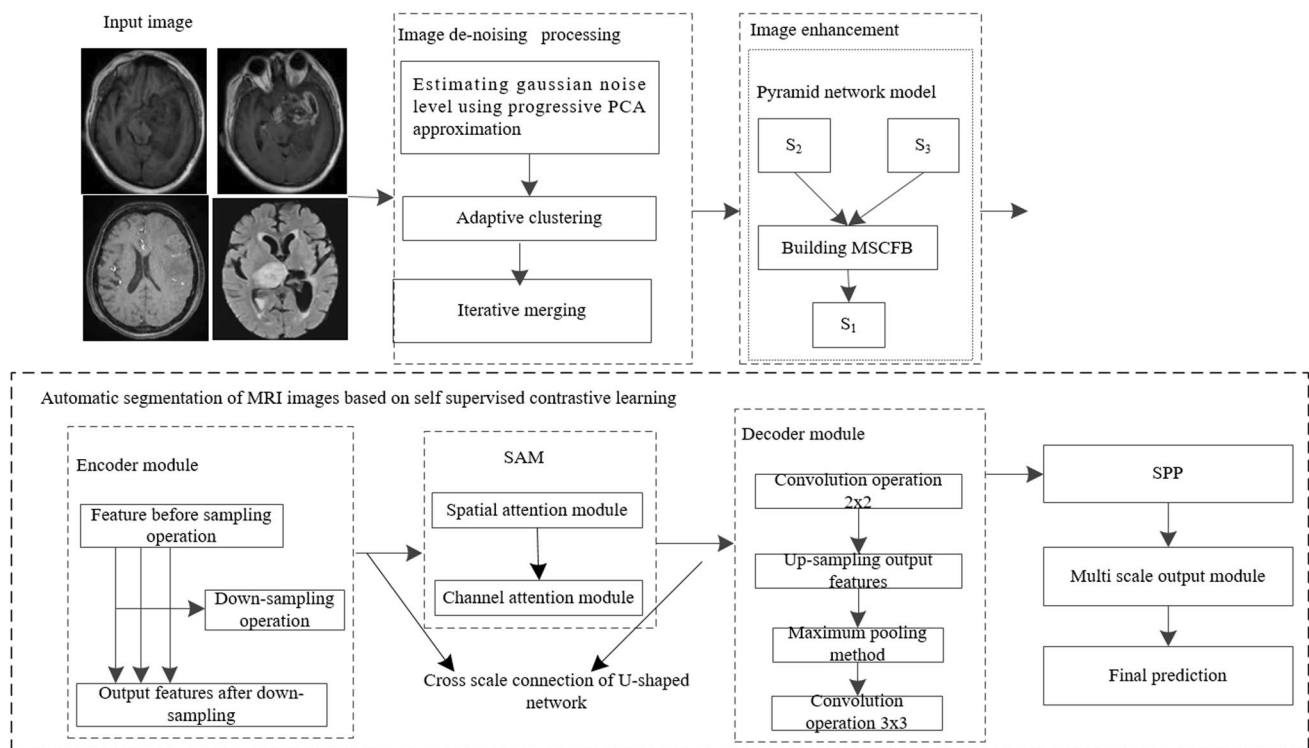


Fig. 1 Framework for this paper

3.1 Data Sets

Two publicly datasets, namely the **OpenfMRI** dataset and **FBIRN** Dataset, were used as experimental datasets. OpenfMRI dataset contains a collection of 50 brain tumor MRI image samples from patients with schizophrenia, while The FBIRN dataset was able to share data from 87 image samples with schizophrenia. The input images of the network were resized to a uniform size of 640×640 . For images with different dimensions, cropping or padding operations can be applied while maintaining the aspect ratio. Cropping can be used to reduce the size of an image, while padding can be used to enlarge it. A total of 80% images were used as the training set, and 20% images were used as the test set. The initial learning rate was 0.003, with a momentum of 0.9 and a weight decay coefficient was 0.001. The Epoch is set to 70.

3.2 Evaluation Metrics

- (1) **Dice coefficient:** the Dice coefficient is a commonly used evaluation metric for image segmentation, used to quantify the accuracy of segmentation results. It measures the degree of overlap between the segmented result and the ground truth. The Dice coefficient ranges from 0 to 1, with values closer to 1 indicating a higher level of overlap between the segmented result and the true

label, representing a better segmentation performance. The formula for calculating the Dice coefficient is as follows:

$$\text{Dice} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}} \quad (1)$$

where TP represents the number of pixels correctly segmented into positive examples (tumor areas); FP represents the number of pixels that were incorrectly divided into positive examples; FN is the number of pixels that were incorrectly divided into negative examples.

- (2) **Recall:** recall is a metric used to evaluate the completeness and coverage of segmentation results. It represents the proportion of pixels correctly classified as positive in relation to the total number of true positive instances. The recall value ranges from 0 to 1, with values closer to 1 indicating that the segmentation results cover more true positive regions, representing a better segmentation performance. The formula for calculating Recall is as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

- (3) **Precision:** precision represents the proportion of pixels correctly classified as positive among all pixels classified as positive. The precision value ranges from 0 to

1, with values closer to 1 indicating higher precision of positive regions in the segmentation result, representing a better segmentation performance. The formula for calculating precision is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (3)$$

- (4) Inference time: it refers to the time required to complete the segmentation of an image. A shorter inference time indicates a faster segmentation algorithm, which is more useful for real-time applications.
- (5) Parameter count: it represents the number of parameters in a segmentation model, providing information about model complexity and storage requirements. A higher parameter count usually indicates a more complex model with potentially higher fitting capacity.
- (6) Model 1 is based on the original UNet encoder–decoder structure. Model 2 utilizes a U-Net network with cross-scale connections. Model 3 combines the original UNet structure with a selective feature fusion attention module. Model 4 incorporates a U-Net network with cross-scale connections and a selective feature fusion attention module. These four models were used for ablation experiments to analyze the impact of network structure on evaluating model performance.

3.3 Progressive Principal Component Analysis (PCA) for Denoising Brain Tumor MRI Images

A denoising algorithm is designed based on progressive PCA approximation and adaptive clustering for the denoising processing of brain tumor MRI images in schizophrenia patients. The specific denoising steps are as follows:

- (1) Global noise level estimation: based on the low-rank assumption and the MP (matching pursuit) theorem, we obtain the relationship between the eigenvalues χ of PCA features and the noise level δ of the random matrix M . Utilizing this relationship, we estimate the Gaussian noise level σ in brain tumor MRI images of patients with schizophrenia.
- (2) Adaptive clustering: after obtaining the global noise level σ , denoising of brain tumor MRI images is achieved by performing a low-rank approximation on a per-class basis. An adaptive clustering method is employed to construct sets of similar image patches for the low-rank matrix.

Design an adaptive clustering method using “over-clustering and iterative merging” with the following steps:

- (1) Set the number of clusters

Set the number of clusters to the following formula:

$$V = \max \left\{ \frac{\alpha\beta}{256^2}, 4 \right\} \quad (4)$$

where α refers to the height of brain tumor MRI images in patients with schizophrenia, while β is width [19].

- (2) Each class is re-clustered to obtain the following equation for the number of clusters

$$n = F \left(\frac{H_{i_0}}{NV\sigma^2} \right) \quad (5)$$

where N denotes the linear minimum mean square error estimation (LMMSE) estimator of local estimated parameters. H_{i_0} refers to the number of block vectors in the i_0 class matrix in the first stage. $F(\cdot)$ is the floor function.

- (3) After performing over-clustering, iterative merging is implemented to avoid clustering that is too small or too dispersed. At this point, a reference value O is set and utilized as the merging threshold.

When the distance between any two classes satisfies the following equation:

$$\|\bar{a} - \bar{b}\|_2^2 < O \quad (6)$$

where $\bar{a}$ and $\bar{b}$ are the mean vector corresponding to two image block groups.

Then the two classes are merged. The iterative process ends when there is no further change in the number of image block groups [20, 21].

- (4) Let $Y_{0,i}$ represent the noise-free class matrix of the i th brain tumor MRI image, and let Y_i represent the noisy class matrix of the i th brain tumor MRI image.

By applying a hard thresholding operation on the Y_i singular values, we achieve the approximation of a low-rank matrix, as shown in equation

$$\tilde{Y}_i = \sqrt{G} \sum_{i=1}^N \sqrt{n\chi_{y,i} Y_{0,i}} \quad (7)$$

where G refers to the number of image block groups. $\chi_{y,i}$ refers to the threshold. $\tilde{Y}_i$ refers to a low rank matrix [22]. The calculation formula is shown in Eq. (5)

$$\chi_{y,i} = \sqrt{\varepsilon\chi} \quad (8)$$

where ε refers to the adjustment coefficient.

The number of K satisfying the following equation is the rank of $\tilde{Y}_i$

$$K > \chi_{y,i} \quad (9)$$

where K represents singular value [23].

3.4 Brain Tumor MRI Image Enhancement Based on Brightness Perception

Design a brightness-aware image enhancement algorithm to perform enhancement processing on brain tumor MRI images [24]. The algorithm process is:

- (1) Input brain tumor MRI image.
- (2) Acquire brain tumor MRI image dataset required for model training and validation.
- (3) Construct a brightness-aware pyramid network model. The model consists of a brightness-aware refinement and reconstruction branch, denoted as S_1 , and two coarse feature extraction branches, denoted as S_2 and S_3 [25]. Global features are obtained in S_2 and S_3 , and passed to S_1 to implement image enhancement.

The input of rough branch S_i is represented by the following equation:

$$\begin{cases} H_{\text{low}}^i = H_{\text{low}}^{i+1} \downarrow \\ i = 2, 3 \end{cases} \quad (10)$$

where $\downarrow$ represents the average pooling operation. H_{low}^{i+1} is the input of another rough branch [26].

The input of branch S_1 is represented by the following equation:

$$H_{\text{low}}^1 = H_{\text{in}} \quad (11)$$

where H_{in} refers to the original input image.

The output and input of each branch are represented by the following equation

$$\begin{cases} H_{\text{high}}^i = \tilde{Y}_i g_{S_i} [g_3(H_{\text{low}}^i), H_{\text{high}}^{i-1} \uparrow] \\ i = 1, 2 \end{cases} \quad (12)$$

where g_{S_i} is the calculation and operation corresponding to branch S_i , while there is no output from the previous layer in S_3 , namely:

$$H_{\text{high}}^3 = g_{S_3}(H_{\text{low}}^3) \quad (13)$$

where g_3 refers to a convolutional layer with a kernel of 3. $\uparrow$ is a deconvolution operation used for image upsampling. H_{high}^{i-1} is the output of the previous branch [27].

- (4) Build an MSCFB, which includes contrast attention mechanism, channel reordering, and channel splitting. Use MSCFB to extract corresponding global features at different scales in the rough feature extraction branch.

In S_1 , an equivalent number of MSCFB need to be introduced to adaptively restore local details of the image.

- (5) Introduce a luminance-aware mechanism in S_1 to adjust brightness by learning the luminance mapping between the target and input images [28]. The luminance-aware mechanism is given by the following expression:

$$j_n = k_{\text{REF}}^n(j_{n-1}) + k_{\text{LUM}}^n(u_{n-1}) \quad (14)$$

where $k_{\text{REF}}^n(\cdot)$ represents the operation of refining the n th module of the branch. j_{n-1} is the output of a MSCFB on a refined branch. $k_{\text{LUM}}^n(\cdot)$ represents the operation of the n th module in the brightness section. u_{n-1} refers to the output of an MSCFB on a rough branch.

- (6) Construct a model loss function, as follows:

$$L_t = L_c + L_l + L_v \quad (15)$$

where L_c refers to content loss. L_l is brightness loss. L_v represents perceptual loss.

3.5 Automatic Segmentation of Brain Tumor MRI Images

Design a cross-scale connected U-Net network with a selective feature fusion attention module based on self-supervised contrastive learning to facilitate automatic segmentation of brain tumor MRI images. First, prepare training data from a large corpus of annotated brain tumor MRI images. Each image includes the original image and its corresponding label (i.e., ground truth segmentation). Additionally, to facilitate self-supervised learning, generate a pair of augmented images, where one image serves as the query image and the other image serves as the positive example. Based on self-supervised contrastive learning methods, it is necessary to define a contrastive loss function to optimize the model. The comparison loss function formula is as follows:

$$L = -\log \frac{\exp\left(\frac{s}{\tau}\right)}{\sum \exp\left(\frac{si}{\tau}\right)} \quad (16)$$

where s represents the similarity measure between the query image and the regular image. τ is a temperature parameter. si represents the similarity measure between the query image and the negative example image. $\sum$ represents the sum of all negative example images. This loss function measures the relative probability of the similarity between the query image and positive example image, as compared to other negative example images. Optimization of this loss function enables the query and positive example images to be closer to each other in the feature space, thereby allowing automatic segmentation of brain tumor MRI images.

A cross-scale connected U-Net network is used as the model architecture, combined with a selective feature fusion attention module. The network has an encoder and decoder section for feature extraction and image restoration, respectively. The selective feature fusion attention module is used to adaptively adjust and fuse features of different scales.

In the aforementioned design, self-supervised learning is conducted by constructing pairs of query and positive images, and using a contrastive loss function to learn the similarity between image features. By continuously optimizing the model parameters, the query and positive images are brought closer together in the feature space, enabling automatic segmentation of brain tumor MRI images. The designed cross-scale connected U-net network adopts the U-net encoder-decoder structure as its base network architecture, implementing only three down-sampling operations to reduce the number of network parameters. Structure of cross scale connected U-Net network is illustrated in Fig. 2.

Figure 2 depicts a typical U-shaped network (U-net) structure. This type of network is commonly used for image segmentation tasks as it can capture contextual information in images while preserving spatial details. The network comprises an encoder on the left and a decoder on the right, connected by cross-scale connections. SAM represents the segmentation attention module. 64, 128, and 256: these numbers represent the channel numbers of the feature maps (i.e., depth). In U-Net, as the network depth increases, the channel numbers of the feature maps gradually increase to capture more diverse feature representations. These numbers indicate the depths of feature maps at different levels. In the encoder, the size of feature maps decreases gradually while the channel numbers increase; in the decoder, the size of feature maps increases gradually, and the channel numbers may reduce accordingly.

In the attention module, cross-layer connections are implemented through skip connections. This module is primarily used for the segmentation of brain tumor MRI images and is placed on all skip connections to achieve efficient connectivity of cross-scale features. This includes increasing the corresponding weights of useful information and decreasing the corresponding weights of irrelevant information.

The skip connection strategy is as follows:

- (1) In the encoding path, the features before down-sampling operations (excluding adjacent features) are connected to the outputs after each down-sampling operation. The encoder is responsible for extracting feature information from the input MRI images and encoding this information into feature maps.
- (2) Place the SAM in each skip connection between the encoder and decoder. The attention module adjusts the importance of features at different scales by increasing the corresponding weights of useful information and decreasing the corresponding weights of irrelevant information, thereby enhancing the accuracy of segmentation.

The design of the SAM is as follows: it consists of a channel attention module and a spatial attention module, wherein the spatial attention module is responsible for generating spatial attention feature maps, while the channel attention module is responsible for generating the final attention feature map.

Given input T_n , attention feature maps are generated using the SAM, as shown in equation

$$R_{att} = d_f(d_r(T_n) \otimes T_n) \otimes T_n \quad (17)$$

where $d_f(\cdot)$ is the generation channel attention feature map function of the channel attention module; $d_r(\cdot)$ refers to the generating spatial attention feature map function of the spatial attention module.

- (3) In the decoding path, the decoder gradually restores the spatial resolution of the image based on the feature information extracted by the encoder and the feature importance adjusted by the attention module. The features after two convolution operations are connected to the outputs after each up-sampling operation. Different scales of max-pooling methods are applied to connect feature maps of varying sizes, with convolution operations of 2×2 and 3×3 to ensure the feature map size consistency across cross-layer connections.
- (4) By using 3×3 convolution operation, features with different scales of connections are given the same number of channels.

The SPP module allows for the extraction of features at multiple scales. The multi-scale output module combines these multi-scale features and serves to accelerate network SPP training and provide deep supervision. Before the final prediction map output, the multi-scale output module obtains three probability maps of the same size as the original image, represented as C_1 , C_2 , and C_3 respectively. The final probability map is obtained by averaging these maps, as shown in equation

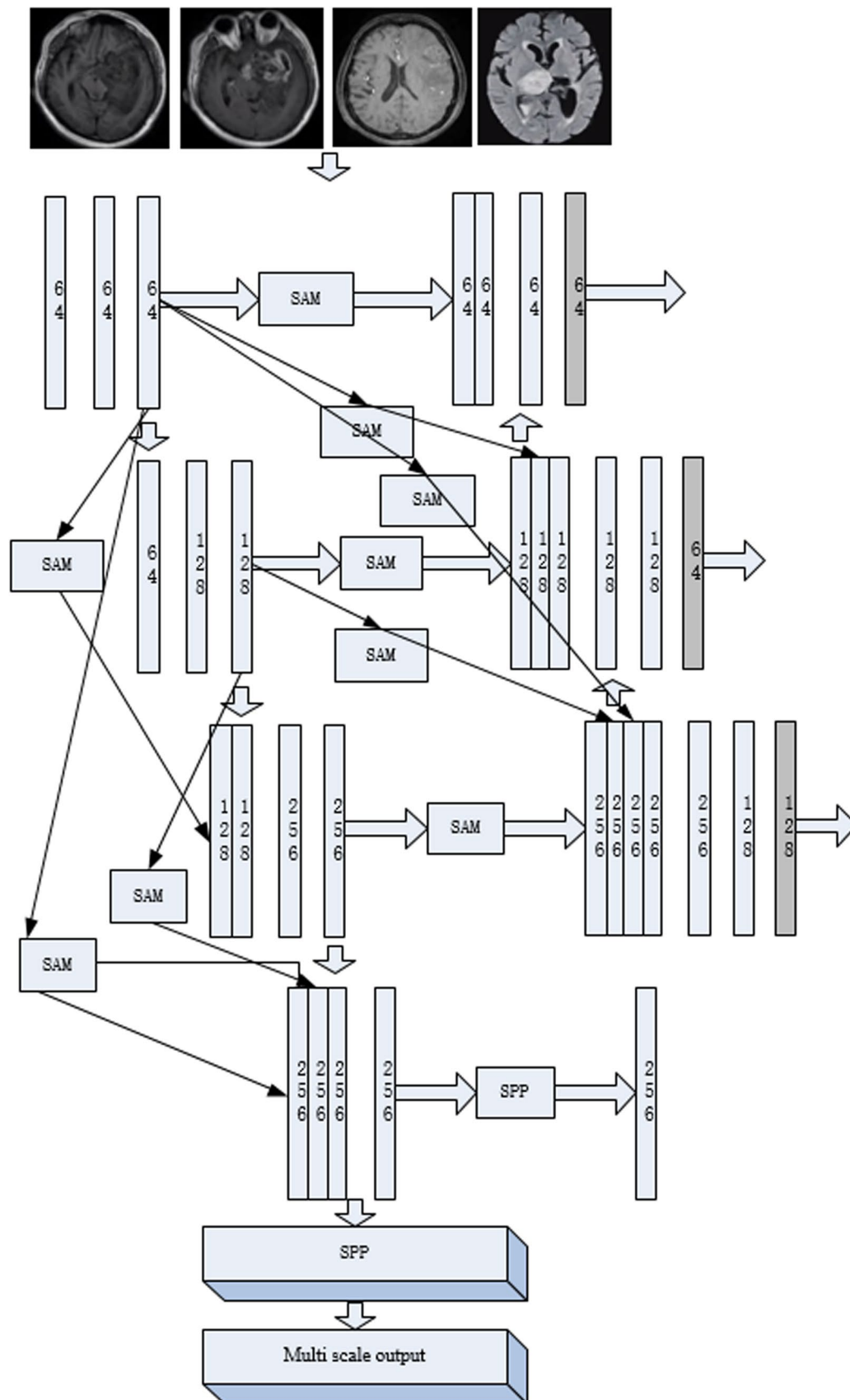


Fig. 2 Structure of cross scale connected U-Net network

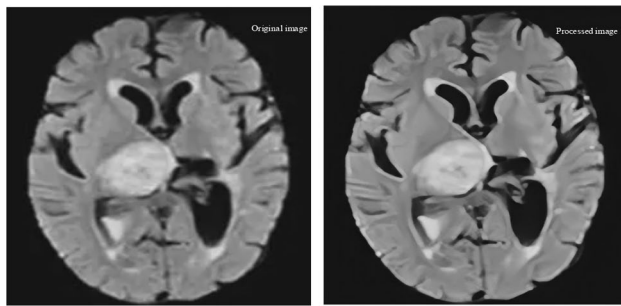


Fig. 3 Comparison of denoised and enhanced images

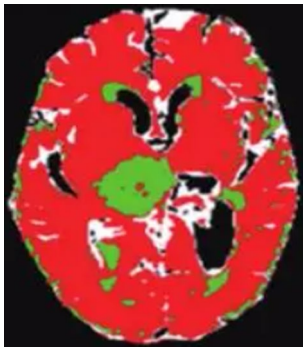


Fig. 4 Automatic segmentation results

$$C = \frac{C_1 + C_2 + C_3}{3} \quad (18)$$

Finally, obtain the final segmentation as shown in formula16.

$$\begin{cases} C > 0.5 \rightarrow 1 \\ \text{else} \rightarrow 0 \end{cases} \quad (19)$$

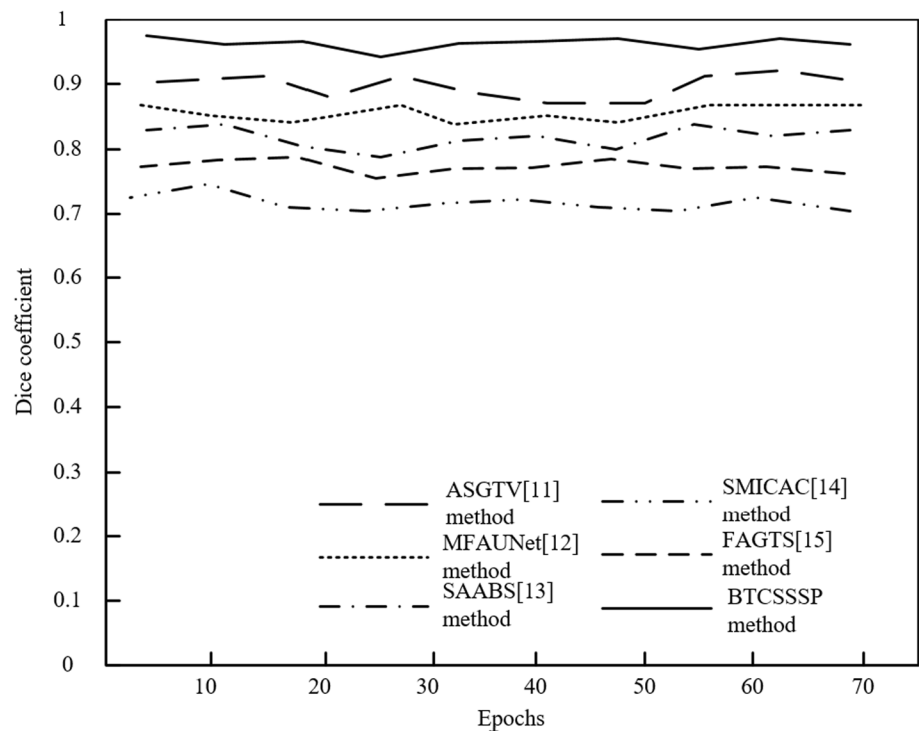
4 Results and Discussion

4.1 Comparison with the State-of-the-Art Methods

Firstly, the experimental dataset is subjected to denoising using a denoising algorithm based on progressive PCA approximation and adaptive clustering. Subsequently, an image enhancement algorithm based on brightness perception is applied to enhance the experimental dataset. The comparison of denoised and enhanced images is shown in Fig. 3.

Finally, the experimental dataset is automatically segmented using a cross-scale connected U-shaped network with a selective feature fusion attention module. The automatic segmentation results are shown in Fig. 4.

Fig. 5 Comparison of Dice coefficient results for six methods



The results of segmenting the brain tumor MRI images of patients with schizophrenia using six different methods can be observed from Fig. 5. The Dice coefficient values for each method are as follows: ASGTV method [11] has a Dice coefficient value of approximately 0.9, MFAUNet method [12] has a Dice coefficient value of approximately 0.85, SAABS method [13] has a Dice coefficient value of approximately 0.8, SMICAC method [14] has a Dice coefficient value of approximately 0.76, FAGTS method [15] has a Dice coefficient value of approximately 0.7, while the BTCSSSP method has a Dice coefficient value of approximately 0.95. Considering these results collectively, the BTCSSSP method performs the best in terms of the Dice coefficient values. The scores close to 1 indicate a high degree of overlap between the segmentation result and the ground truth labels, demonstrating excellent segmentation performance. This advantage can be attributed to the adoption of self-supervised contrastive learning, which helps enhance the learning capability of the network. Through self-supervised contrastive learning, the network generates automatic pairs of positive and negative samples and learns feature representations by encouraging the network to bring similar samples closer and dissimilar samples farther apart. Consequently, the BTCSSSP method can capture the features of the brain tumor region better, thereby improving the precision of the segmentation results. Furthermore, by comparing the results of other methods, it can be observed that the Dice coefficient values gradually decrease. This may be due to the failure of these methods to consider the special circumstances of patients with schizophrenia in brain tumor segmentation or their ineffective utilization of crucial information in the images, leading to lower segmentation precision.

According to Table 1, we can see that the maximum recall values for ASGTV method [11], MFAUNet method [12], SAABS method [13], SMICAC method [14], and FAGTS method [15] are 0.85, 0.84, 0.88, 0.81, and 0.85, respectively. The BTCSSSP method has a maximum recall value of 0.95, which is higher than the recall values of the compared methods. This indicates that the BTCSSSP method

has better segmentation performance. The improved performance of the BTCSSSP method can be attributed to the introduction of the Selective Feature Fusion Attention Module. This module dynamically adjusts the network's focus on features of different scales. By fusing features and controlling attention, the network can better capture tumor regions of different sizes and shapes, thus improving the recall of detection and segmentation. In summary, the BTCSSSP method outperforms the compared methods in terms of recall. The introduced Selective Feature Fusion Attention Module enables the network to effectively capture tumor regions of different scales and shapes, leading to improved segmentation precision and recall. The compared methods might not consider the special characteristics of brain tumor segmentation in patients with schizophrenia or effectively utilize key information in the images, resulting in lower segmentation precision.

According to Fig. 6, it can be seen that the precision of image segmentation for ASGTV method [11] is approximately 0.89, MFAUNet method [12] is approximately 0.75, SAABS method [13] is approximately 0.72, SMICAC method [14] is approximately 0.69, and FAGTS method [15] is approximately 0.65. In contrast, the BTCSSSP method achieves a precision of approximately 0.95 for image segmentation. The results clearly indicate that the BTCSSSP method outperforms the compared methods in terms of image segmentation. This is because the BTCSSSP method introduces a Selective Feature Fusion Attention Module, which dynamically adjusts the network's attention to features of different scales. This allows the network to better capture the characteristic information of brain tumors at various scales, resulting in more accurate segmentation results. In conclusion, the BTCSSSP method demonstrates significantly higher precision in image segmentation compared to the compared methods. By incorporating the Selective Feature Fusion Attention Module, the BTCSSSP method enables the network to effectively capture features at different scales, thereby improving the precision of segmentation results.

Table 1 Comparison of recall results of six methods (%)

Epochs	ASGTV [11] method	MFAUNet [12] method	SAABS [13] method	SMICAC [14] method	FAGTS [15] method	BTCSSSP method
10	0.82	0.81	0.84	0.79	0.86	0.92
20	0.85	0.80	0.85	0.78	0.89	0.91
30	0.84	0.83	0.83	0.80	0.85	0.89
40	0.80	0.79	0.87	0.76	0.84	0.93
50	0.79	0.84	0.88	0.75	0.84	0.95
60	0.83	0.80	0.86	0.77	0.82	0.90
70	0.81	0.81	0.84	0.81	0.87	0.93

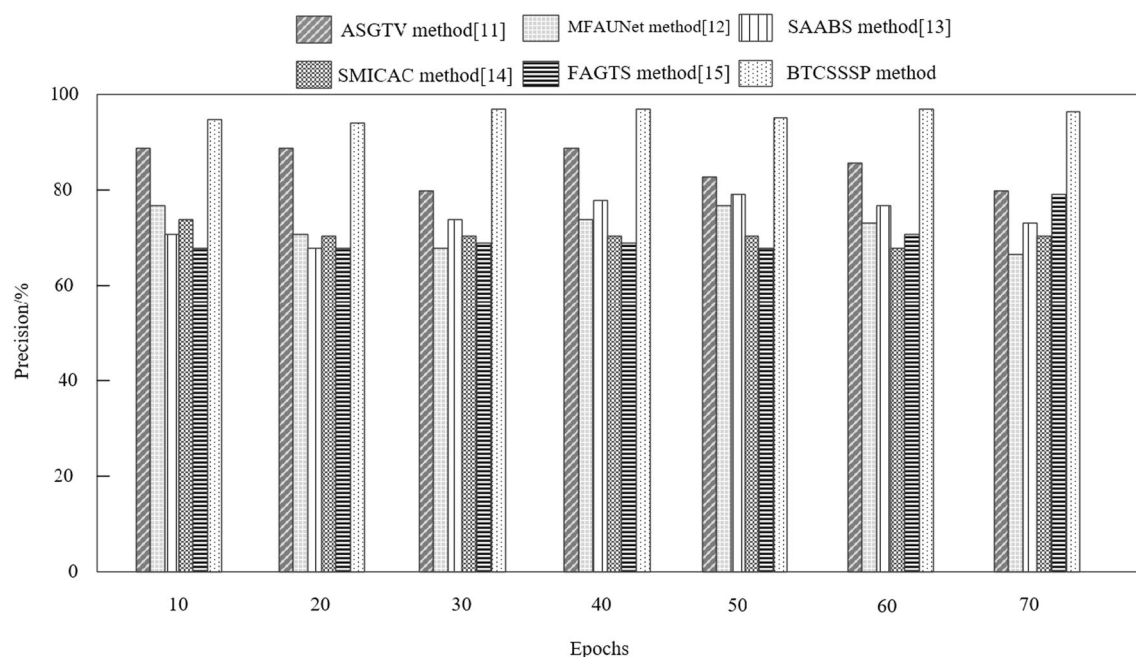


Fig. 6 Precision comparison results of six methods

Table 2 Comparison results of inference time (ms)

Epochs	ASGTV [11] method	MFAUNet [12] method	SAABS [13] method	SMICAC [14] method	FAGTS [15] method	BTCSSSP method
10	368	350	320	405	330	298
20	375	365	335	411	339	305
30	388	374	345	435	342	310
40	394	384	357	454	349	324
50	402	394	364	460	350	330
60	425	400	378	464	358	340
70	430	435	380	468	360	345

Based on the data from Table 2, we can observe that the segmentation time for all six methods increases as the number of samples increases. Specifically, ASGTV method [11] has a segmentation time ranging from 368 to 430 ms, MFAUNet method [12] is from 350 to 435 ms, SAABS method [13] is from 320 to 380 ms, SMICAC method [14] is from 405 to 468 ms, FAGTS method [15] is from 330 to 360 ms, and the BTCSSSP method has a segmentation time ranging from 298 to 345 ms. Importantly, the segmentation time of the BTCSSSP method is consistently lower than that of the compared methods, indicating a higher segmentation efficiency. This is because the BTCSSSP method utilizes a U-Net

network structure with cross-scale connections. This network structure enables multi-level feature extraction and fusion on the image. By connecting features across different scales, the network can simultaneously leverage low-level and high-level features, thereby fully mining information within the image. Specifically, the BTCSSSP method utilizes this design to learn the tumor features more efficiently, reducing the need for redundant computations and ultimately lowering the segmentation time. In conclusion, through comparative analysis, we can conclude that the segmentation time of the BTCSSSP method is consistently lower than that of the compared methods. This is because the BTCSSSP method's network

Table 3 Comparison results of parameter count (piece)

Epochs	ASGTV [11] method	MFAUNet [12] method	SAABS [13] method	SMICAC [14] method	FAGTS [15] method	BTC- SSSP method
10	301	252	258	350	220	156
20	305	254	254	345	231	159
30	315	255	259	352	228	160
40	324	258	264	364	224	158
50	314	254	261	321	234	156
60	315	258	259	346	246	164
70	321	260	270	359	247	160

structure can more efficiently learn tumor features and reduce the need for redundant computations. The results of this study provide a faster and more efficient method for brain tumor segmentation.

Based on the data in Table 3, we can see that the minimum parameter count of ASGTV method [11] is 301, MFAUNet method [12] is 252, SAABS method [13] is 254, SMICAC method [14] is 321, FAGTS method [15] is 220, and the BTCSSSP method has a minimum parameter count of 156. It is important to note that the BTCSSSP method has a lower parameter count compared to the other methods, indicating a faster segmentation rate. This is because the BTCSSSP method optimizes the parameter settings through network structure design, self-supervised learning, and the application of attention mechanisms, among other techniques. By incorporating these approaches, the BTCSSSP method achieves a more efficient use of parameters, resulting in a reduction in the total parameter count. In summary, based on the analysis of parameter counts, we can conclude that the BTCSSSP method has a lower parameter count compared to the other methods. This indicates a faster segmentation rate, as the optimization of parameter settings through various techniques allows for a more efficient use of parameters in the BTCSSSP method.

4.2 Ablation Study

According to the results of the ablation experiments in Table 4, Model 2 achieved a Dice score of 0.83, recall of

73.25%, precision of 82.32%, with a parameter count of 332 and an inference time of 50 ms. On the other hand, the Model 1 achieved a Dice score of 0.82, recall of 69.32%, precision of 78.52%, with a parameter count of 356 and an inference time of 562 ms. After incorporating the selective feature fusion attention module, all performance metrics of the original network were improved. By replacing Model 2 with Model1, and combining it with the selective feature fusion attention module (Model 3), the highest test values for Dice score, recall, and precision were achieved. Additionally, the parameter count and inference time were also significantly reduced, indicating that the designed model exhibits improved performance. In summary, the results demonstrate that the BTCSSSP method performs well. The U-Net architecture with cross-scale connections, along with the selective feature fusion attention module (Model 4), improves segmentation performance while reducing the number of parameters and inference time.

5 Conclusions

Unhealthy modern lifestyles may lead to the development of rare but highly harmful diseases, such as brain tumors. Accurate segmentation of brain tumors in patients with schizophrenia is crucial for treatment and monitoring purposes. To address the issues of excessive parameters and time-consuming processes in traditional methods, we propose an automatic segmentation method for brain tumors in schizophrenia patients using

Table 4 Comparison results of ablation experiment

Model	Dice coefficient	Recall (%)	Precision (%)	Parameter count (pieces)	Inference time (ms)
Model 1	0.82	69.32	78.52	356	562
Model 2	0.83	73.25	82.32	332	504
Model 3	0.84	75.20	84.25	302	486
Model 4	0.96	91.25	95.20	156	298

self-supervised contrastive learning. Our method first employs progressive PCA approximation and adaptive clustering algorithms to reduce noise interference in the images, effectively improving the image quality. Then, we utilize a brightness-aware image enhancement algorithm to address issues such as image inhomogeneity, blurry boundaries, and low spatial resolution, resulting in clearer and more distinct delineation of the tumor regions. Finally, we introduce a self-supervised contrastive learning-based U-net with multiscale connectivity and a selective feature fusion attention module, enabling automatic tumor segmentation. This method contributes to more accurate diagnosis and treatment of brain tumors in patients with schizophrenia and can be applied for clinical-assisted diagnostics and treatment, providing doctors with more precise and efficient decision-making support. However, there are still some potential limitations in the study. For example, although the number of model parameters has been reduced, the complex network structure and attention mechanisms may increase computational complexity, affecting efficiency in practical applications. While the denoising and image enhancement algorithms improve image quality, they may also lead to the loss of some important details. Future works include improving the multiscale connectivity U-net to better handle small-scale structures and details, enhancing the precision and sensitivity towards edge and texture information. Additionally, we will explore the design of more effective mechanisms for modeling and segmenting large-scale tumor structures to improve the segmentation capability and coverage of the entire tumor region.

Appendix

Abbreviations of this paper are shown in Table 5.

Table 5 Abbreviations of nomenclature of this paper

Nomenclature	Abbreviation
Magnetic resonance imaging	MRI
convolutional neural networks	CNN
Principal component analysis	PCA
linear minimum mean square error estimation	LMMSE
Multi-Scale Context Feature Blocks	MSCFB
Segmentation attention module	SAM
Spatial pyramid pooling	SPP
Matching Pursuit	MP

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Data Availability The data used to support the findings of this study are included in the article.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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